

**Hyunseok Seung**  
1205 University Avenue  
Madison, WI 53706  
Email: [hseung2@wisc.edu](mailto:hseung2@wisc.edu) | Website: [hseung88.github.io](https://hseung88.github.io)

## Professional Appointments

2025 – Present      **Postdoctoral Research Associate**, *Department of Statistics, University of Wisconsin–Madison*, Madison, WI  
Advisor: [Matthias Katzfuss](#)

## Education

2025      **Ph.D.** in Statistics, *University of Georgia*, Athens, GA  
Advisors: [Jaewoo Lee](#) and [Yuan Ke](#)  
Dissertation: *Scalable and Efficient Learning: Algorithmic Advances for Time Series and Deep Neural Models*

2018      **M.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea  
Advisor: [Sangun Park](#)  
Thesis: *Modified Likelihood Ratio Tests for Extreme Value Distributions*

2016      **B.A.** in Applied Statistics, *Yonsei University*, Seoul, South Korea

## Publications

### Peer-reviewed Conference Proceedings

- C1. **Hyunseok Seung**, Lee, J. & Ko, H. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* in *AAAI Conference on Artificial Intelligence* (2026).
- C2. **Hyunseok Seung**, Lee, J. & Ko, H. *MAC: An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* in *IEEE International Conference on Data Mining (ICDM)* (2025).
- C3. **Hyunseok Seung**, Lee, J. & Ko, H. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* in *IEEE International Conference on Data Mining Workshop (ICDMW)* (2024).
- C4. **Hyunseok Seung**, Lee, J. & Ko, H. *NysAct: A Scalable Preconditioned Gradient Descent using Nystrom Approximation* in *IEEE International Conference on Big Data* (2024).

### Journal Articles

- J1. **Hyunseok Seung**, Lee, J. & Ko, H. *Mean Activation Curvature for Scalable Second-Order Optimization in Deep Networks*. *Knowledge and Information Systems*. (Invited journal extension from the IEEE ICDM 2025 best-ranked paper selection) (2026).
- J2. **Hyunseok Seung** & Park, S. *Modified Likelihood Ratio Tests for Extreme Value Distributions*. *Communications in Statistics–Theory and Methods* **52**, 5742–5751 (2023).

## Under Review

W1. **Hyunseok Seung**, Han, K., Shen, Y. & Ke, Y. *Enhancing COVID-19 Mortality Prediction with Online Autocovariance Change Points Detection*

## Research

- 2025 – Present      **Postdoctoral Researcher**, *Department of Statistics, University of Wisconsin–Madison*
- **Gaussian Processes with Derivatives** (advised by [Matthias Katzfuss](#))
    - Developed a scalable derivative Gaussian process inference method that replaces full gradient conditioning with low-dimensional gradient reduction while preserving the exact gradient information required for target-specific prediction.
  - **Hyperspectral Foundation Models** (collaboration with [Townsend Lab](#))
    - Pre-training vision transformer foundation models on hyperspectral data, followed by fine-tuning for downstream trait prediction. Conformal prediction to quantify uncertainty.
- 2022 – 2025      **Research Assistant**, *School of Computing, University of Georgia*
- **Deep Learning Optimization** (advised by [Jaewoo Lee](#))
    - Developed a curvature-aware zeroth-order optimization method for fine-tuning LLMs, achieving faster convergence and higher test accuracy than state-of-the-art methods, while cutting memory usage by up to 27% compared to MeZO-Adam.
    - Developed scalable second-order optimization methods using activation covariance, improving test accuracy by 3.6% on vision transformers compared to AdamW.
- 2023 – 2024      **Research Assistant**, *Department of Educational Psychology, University of Georgia*
- **Topic Modeling** (advised by [Shiyu Wang](#))
    - Analyzed video and text data using automatic speech recognition and topic modeling, collaborating with researchers in mathematics education and psychology.
- 2021 – 2023      **Research Assistant**, *Department of Statistics, University of Georgia*
- **Time Series Forecasting** (advised by [Yuan Ke](#))
    - Developed hybrid COVID-19 mortality forecasting models. Utilized online autocovariance change point detection to boost model accuracy by 6% and reduce training time by 99% compared to standard rolling-window cross validation.
- 2018      **Associate Researcher**, *SK hynix Inc.*
- **Wafer Failure Early Detection System** (advised by [Sangun Park](#))
    - Streamlined semiconductor production by identifying key predictors of wafer failure, using statistical models for high-dimensional fabrication data.
- 2017 – 2018      **Research Assistant**, *Department of Applied Statistics, Yonsei University*
- **Modified Likelihood Ratio Tests** (advised by [Sangun Park](#))
    - Developed modified likelihood ratio test statistics tailored to highly skewed settings to improve sensitivity to tail departures.

## Awards & Honors

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| 2019 – 2024 | Teaching Assistantship (Stipend), <i>University of Georgia</i>                    |
| 2018        | Industry-Sponsored Scholarship Recipient, <i>SK hynix Inc.</i>                    |
| 2018        | Best Paper Presentation Award, <i>Korean Data and Information Science Society</i> |
| 2017 – 2018 | Brain Korea 21 Fellowship, <i>Korean Government-Funded Graduate Program</i>       |
| 2015        | Honors Award, <i>Yonsei University</i>                                            |

## Conference Presentations

### Oral

- T1. *An Efficient Gradient Preconditioning using Mean Activation Approximated Curvature* 2025 IEEE International Conference on Data Mining (Washington, DC, USA). Nov. 2025.
- T2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* 2024 IEEE International Conference on Big Data (Washington, DC, USA). Dec. 2024.
- T3. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Data and Information Science Society (Pukyong University, Busan, South Korea). May 2018.

### Posters

- P1. *Low-Rank Curvature for Zeroth-Order Optimization in LLM Fine-Tuning* AAAI Conference on Artificial Intelligence (Singapore). Jan. 2026.
- P2. *A Scalable Preconditioned Gradient Descent using Nystrom Approximation* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2025.
- P3. *An Adaptive Method Stabilizing Activations for Enhanced Generalization* AI Research Day, Institute for Artificial Intelligence (Athens, GA, USA). Apr. 2024.
- P4. *Modified Likelihood Ratio Tests for Extreme Value Distributions* The Korean Statistical Society (Pusan National University, Busan, South Korea). May 2018.

## Teaching

### University of Georgia

|             |                                                            |             |
|-------------|------------------------------------------------------------|-------------|
| 2019 – 2023 | <b>Teaching Assistant</b>                                  |             |
|             | – STAT6430 Design Analysis Experiments                     | Spring 2023 |
|             | – STAT6315 Statistical Methods for Researchers             | Spring 2023 |
|             | – STAT8330 Advanced Statistical Applications and Computing | Fall 2022   |
|             | – STAT6420 Applied Linear Models                           | Fall 2022   |
|             | – STAT4230 Applied Regression Analysis                     | Spring 2022 |
|             | – STAT2360 Program and Data Lit using R,                   | Fall 2021   |
|             | – STAT4210 Statistical Method,                             | Spring 2021 |
|             | – STAT8440 Statistical Inference Bioinformatics            | Fall 2020   |
|             | – STAT6210 Intro to Statistical Methods                    | Fall 2020   |
|             | – STAT3110 Intro to Statistics for Life Science            | Summer 2020 |

- STAT3120 Intro to Probability for Life Science
- STAT6210 Statistical Methods

Spring 2020  
Fall 2019

## Yonsei University

|             |                                       |             |
|-------------|---------------------------------------|-------------|
| 2018        | <b>Lecturer</b>                       |             |
|             | – STAT1001 Introduction to Statistics | Fall 2018   |
| 2017 – 2018 | <b>Teaching Assistant</b>             |             |
|             | – STAT1001 Introduction to Statistics | Spring 2018 |
|             | – STAT1001 Introduction to Statistics | Fall 2017   |
|             | – STAT1001 Introduction to Statistics | Spring 2017 |

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